
Data Engineering Course

Pandas: Column Selection, Row Selection, Boolean Indexing & Sorting

ASSIGNMENT

Instructor: Tanishq Tyagi

Practice Assignment — Pandas DataFrame Operations

Student Instructions

Read each question carefully and write Python code using the pandas library to produce the required output. All questions are based on a single DataFrame named **students**, which is provided below. Use appropriate pandas techniques covered in class, including column selection, row selection using **loc** and **iloc**, boolean indexing (filtering), and sorting. Attempt every question in the order given, and ensure your code runs without errors before submission.

Dataset Used in This Assignment

All questions below operate on the following pandas DataFrame. Use this exact dataset while solving each question.

```
import pandas as pd

students = pd.DataFrame({
    "Student_ID": [1, 2, 3, 4, 5, 6, 7, 8],
    "Name": ["Aman", "Riya", "Karan", "Neha", "Ankit", "Priya", "Rohan", "Simran"],
    "Course": ["Python", "Java", "Python", "SQL", "Java", "Python", "SQL", "Java"],
    "Age": [21, 20, 22, 23, 21, 24, 22, 20],
    "Marks": [88, 76, 91, 69, 82, 95, 73, 85],
    "City": ["Delhi", "Mumbai", "Delhi", "Pune", "Delhi", "Mumbai", "Bangalore", "Delhi"]
})
```

Assignment Questions

Section A: Column Selection

1. Display only the Name column.
2. Display the Name, Course, and Marks columns.
3. Display all columns except Student_ID.

Section B: Row Selection (loc & iloc)

4. Display the first 5 students using iloc.
5. Display rows 2 to 6 using loc.
6. Display rows 1 to 4 with only the Name, Marks, and City columns using loc.

Section C: Boolean Indexing (Filtering)

7. Display students scoring more than 80 marks.
8. Display students enrolled in the Python course.
9. Display students from Delhi.
10. Display students from Delhi who scored more than 85 marks.
11. Display students whose Age is greater than 21.
12. Display students enrolled in either Java or SQL.
13. Display students not enrolled in the Python course.

Section D: Sorting Data

14. Sort the DataFrame by Marks (Highest to Lowest).
15. Sort the DataFrame by Course and then by Marks.
16. Sort the DataFrame by City and then by Name.

Section E: Combined Application

17. Create a report containing only Name, Course, Marks, and City for Python students scoring 90 or above, sorted by Marks (Highest to Lowest), and reset the index.